



Robo-dogs enforce lockdown

Robot dogs are patrolling Shanghai to help enforce a lockdown curbing the spread of the virus. <https://digit.in/apr22-11>



Nanotech to speed up research

DNA nanotechnology could speed up pharmaceutical development while minimising costs. <https://digit.in/apr22-12>



BIOMETRICS

Sing this song to unlock your phone

Satvik Pandey | feedback@digit.in

Like most technologies in the modern-day, biometrics too has come a long way to become what it is today. From a technological innovation that was strictly reserved for use by a small group of people to being a part of our smartphones, helping us unlock them hundreds of times a day, biometrics have truly become an integral part of our daily lives.

The word biometrics finds its origins in Greece, 'bio' meaning 'life' and 'metry' meaning 'to measure'. But it wasn't until the 1980s that this word shot into the mainstream and started being used to describe the techniques which we know of today. As stated in the research paper titled '*The Historical Development of Biometric Authentication Techniques*', the first reference of this word can be found in a New York

Times article published in 1981.

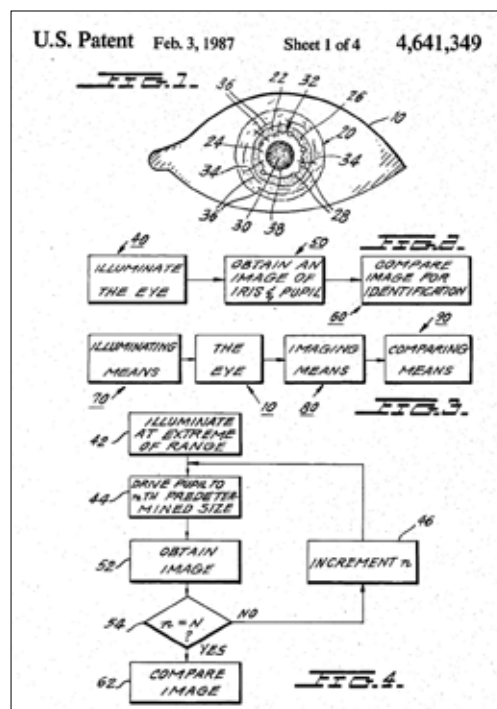
If we trace the history of biometrics, we will land in the world of Assyrians, Babylonians, Japanese, and the Chinese, as they were the ones to have used fingerprints for the signing of legal documents and verifying the legality of the transaction(s) in question. In the modern world, closer to the way we know it, biometrics make the first mark in history around the late 1800s. But the story is a bit controversial.

In a popular journal of the time called '*Nature*', there were a bunch of articles and their counters published, which debated about who would be credited to have devised or discovered the modern technique of biometric authentication. There were two articles that are most widely acknowledged. The first one is by Henry Faulds, who was a Scottish medical missionary in Tokyo, that talked about Henry's observations about fingerprints and how they

could be visually inspected to identify potential criminals. In response to this, Sir William Herschel, a British civil servant working in India, published an article where he highlighted his application of fingerprint identification techniques, which he had been using for the identification of the British subjects in the region.

It took another two decades for this system of identification to be officially established. It wasn't until 1901 that the first British fingerprint files came to being in London, marking the start of the authorised use of biometrics. A couple of years later, in 1903, the next stage of development took place in a city in New York. The New York State Prison System began the systematic use of fingerprints for the identification of criminals. The following year saw the establishment of one of the first fingerprint bureaus by the United States Penitentiary at Leavenworth, Kansas, and the St. Louis, Missouri, Police Department.

Fast forward a couple of decades, and the next advancement took place in the year 1936. An ophthalmologist by the name of Frank Burch proposed



The official drawings submitted for the patent of the iris recognition system